The Credit Concentration Channel

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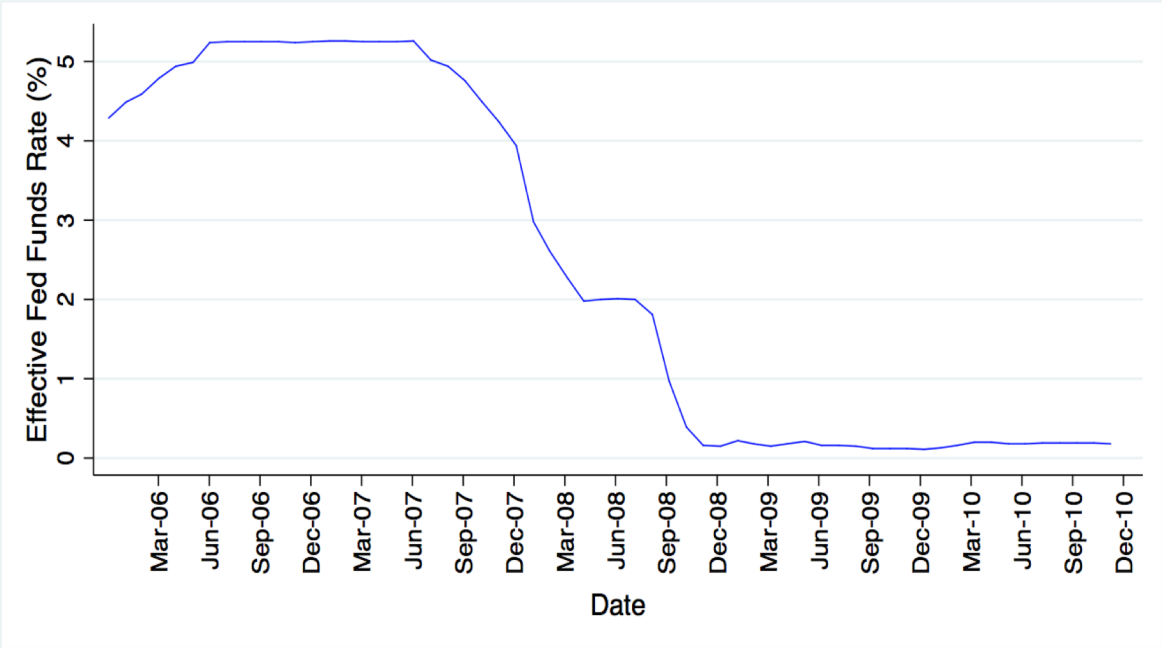
Motivation

- Aggressive monetary easing during the 2007-2008 financial crisis in the U.S.
- Policy concern: Frictions may affect the effectiveness of transmission



The Lending Channel of Monetary Policy Transmission

- Monetary policy transmits to the economy via the "lending channel" (*Bernanke and Blinder (1992), Kashyap and Stein (2000)*)

- Lenders transmit changes in policy rates to borrowers via:
  - Changes in the price of credit supplied (+ve relation)
  - Changes in the quantity of credit supplied (-ve relation)
  - This paper's focus: **the quantity of credit**

- Cross-sectional differences in the effectiveness of transmission via:
  - Differences in the financial condition of lenders
  - Differences in local market characteristics
  - This paper's focus: **credit concentration**

Research Question and Contribution

- Research Question
  - Does credit concentration dampen the response of credit supply to monetary easing?

- Contribution
  - Bring more precise measures of credit supply and credit concentration to this literature
  - Provide micro level evidence of:
    1. The existence of the credit concentration channel
    2. Its ubiquitous influence

Results

- Existence of the credit concentration channel
  1. Lenders' credit supply was less responsive to monetary easing in counties with higher credit concentration
  2. Within counties, lenders with higher credit market power were the least responsive
  3. This channel is separate from the deposit channel (*Drechsler, Savov, and Schnabl, 2017*) and accounts for about 10% of the transmission

- Ubiquitous influence of the credit concentration channel
  1. **Across products**: Affects both new loans and refinances
  2. **Across lenders**: Operates through depository and non-depository lenders, market leaders and laggards
  3. **Across markets**: Is present in markets that were the least hit and those that were the hardest hit by declines in housing prices

Empirical Challenges

CHALLENGE	DESCRIPTION	SOLUTION
Measurement	Need to measure credit supply and credit concentration appropriately	I use mortgage loan application data from the Home Mortgage Disclosure Act (HMDA)*

* HMDA is a close-census of mortgage activity in the United States that includes information about loan/borrower characteristics, the location of the property, and the identity of the lender

CHALLENGE	DESCRIPTION	SOLUTION
Estimation	Need a monetary policy shock to relate credit concentration to the response of credit supply	I use the monetary easing of 2008

Measuring Credit Supply

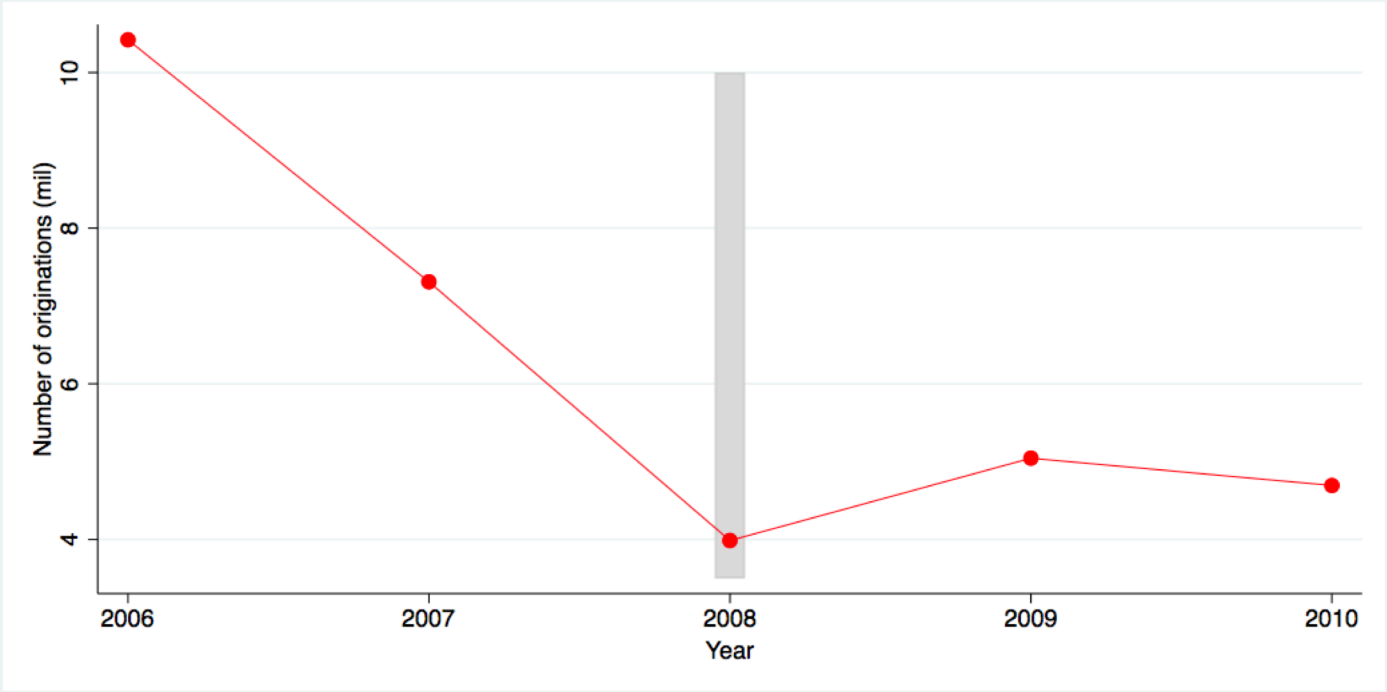
- The "perfect" measure would capture:
  - The <u>total</u> supply of credit
  - As <u>separate</u> from credit demand

- Measures typically employed in the literature:
  - Change in loans held on balance sheet: *Olivero, Li, and Jeon (2011)*
    - Incomplete, particularly so for mortgage credit
    - Can be driven by demand
  - Volume of loan originations: *Scharfstein and Sunderam (2016)*
    - Fares well on completeness
    - Can be driven by demand

- <u>I measure credit supply as the proportion of loan applications that resulted in origination</u>

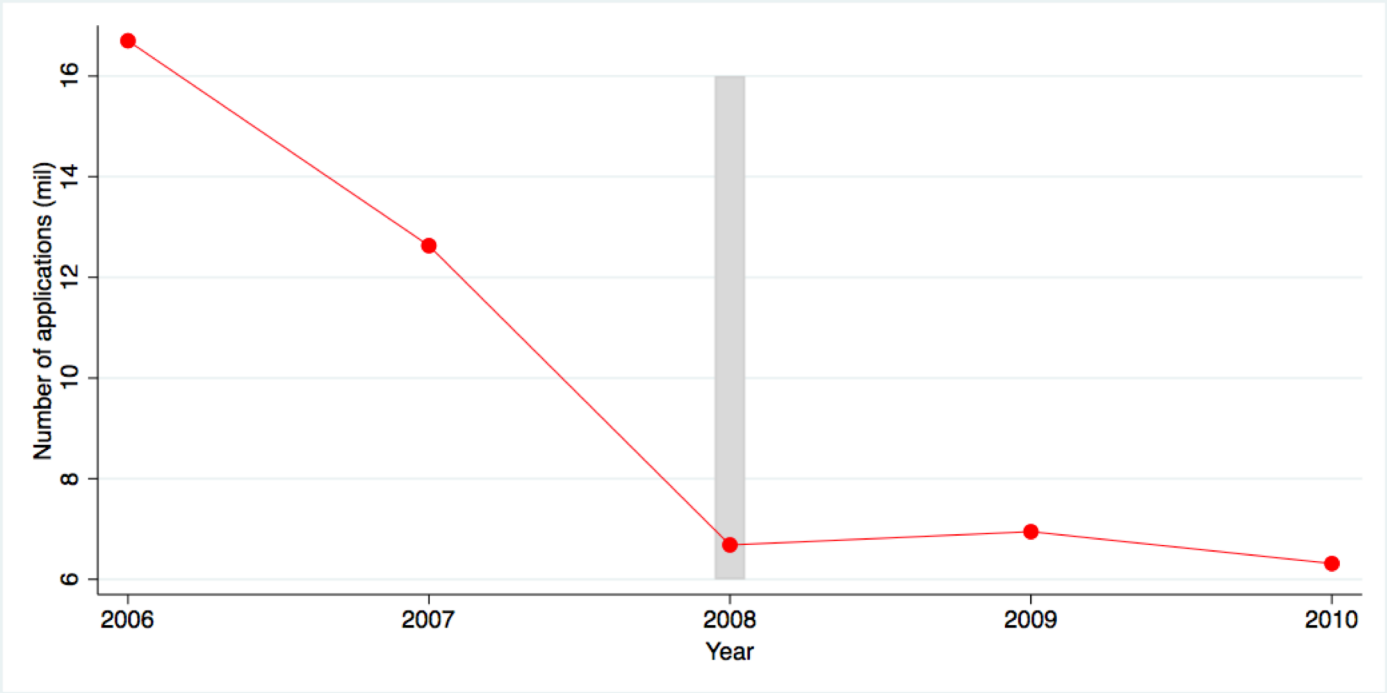
Credit Activity from Loan Originations

- The number of mortgage loan originations declines dramatically
  - Demand or supply?



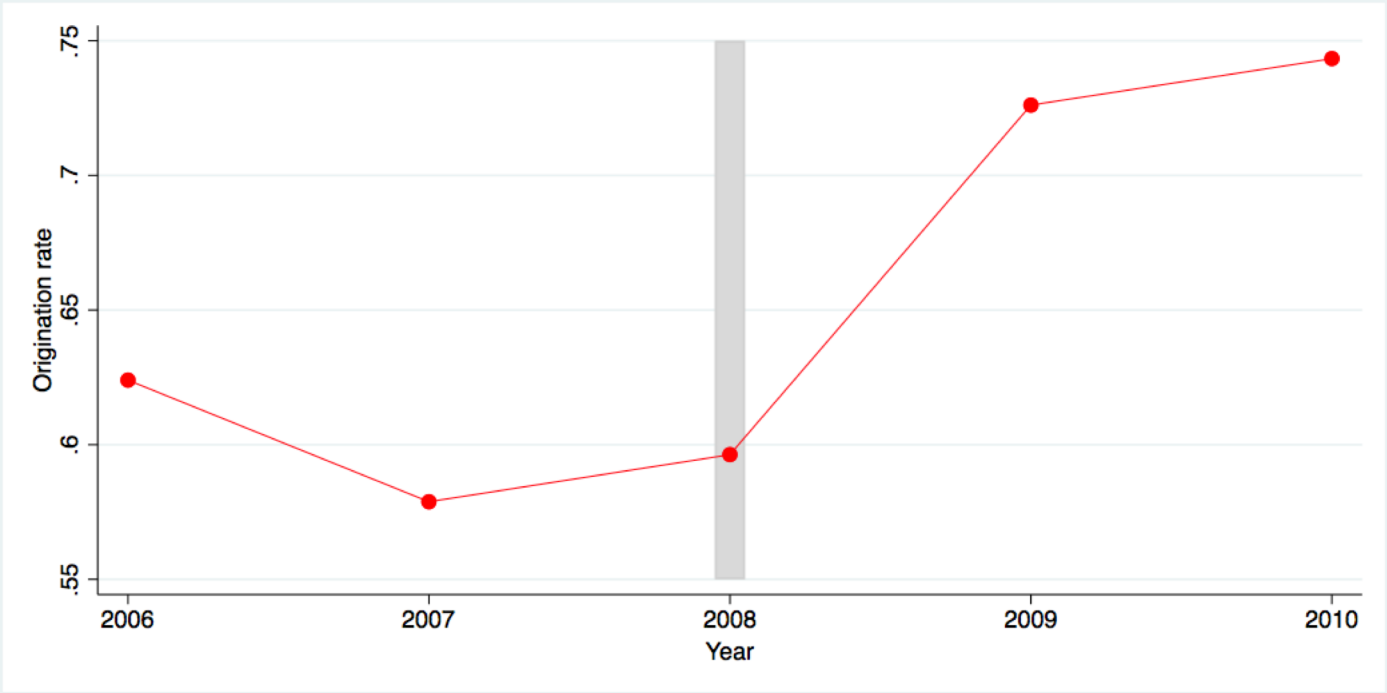
Demand Component: Number of Loan Applications

- Originations (previous graph) drop in sync with applications (this graph)
  - i.e., demand is a major driver



Supply Component: Loan Origination Rate

- Supply responds sharply to monetary easing



Measuring Credit Concentration

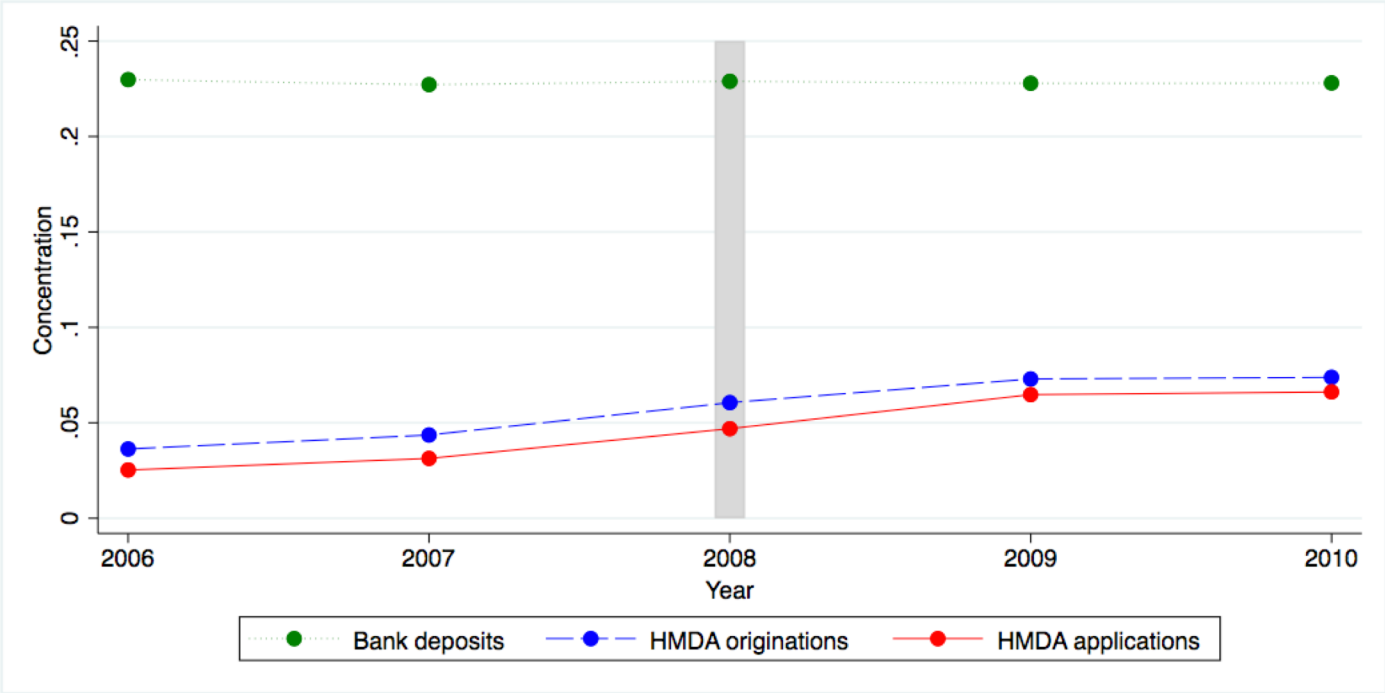
- Which concentration index?
  - Various indices that use **market shares** as the input
  - <u>I use the Herfindahl-Hirschman index:</u> $HHI = \sum_{i=1}^{n} Share_i^2$

- Which market?
  - Mortgage markets are local (Amel, Kennickell and Moore (2008))
  - <u>I define the local market at the county level</u>

- Which share?
  - Deposit shares: *Adams and Amel (2011), Drechsler, Savov, and Schnabl (2017)*
    - Imperfect mapping to lending, effect aggregates up to the lender level
  - Share of on-balance sheet loans: *Van Leuvensteijn et al (2013)*
    - Misses off-balance sheet activity, a stock variable may not reflect the present
  - Share of loan originations: *Scharfstein and Sunderam (2016)*
    - Measure can correlate with credit supply
  - <u>I use the share of mortgage loan applications</u>

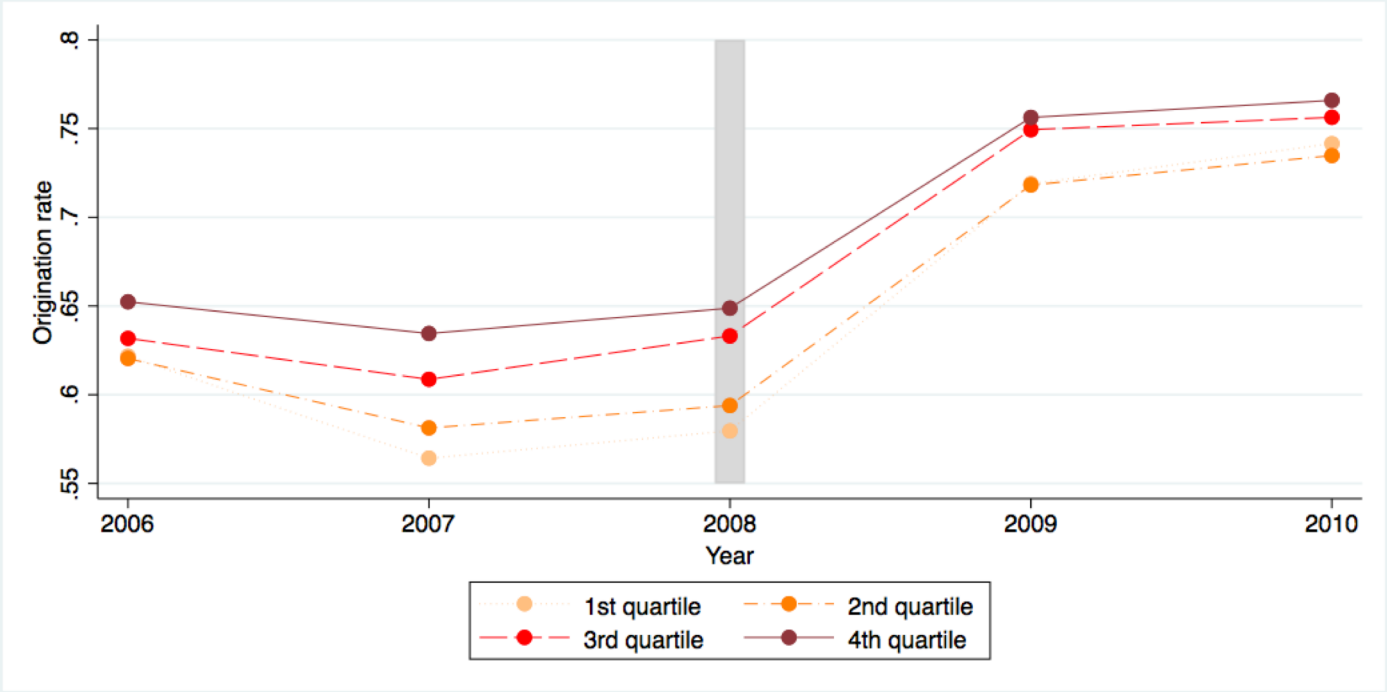
Market Concentration: Different Market Shares as Inputs

- HHI using market shares from bank deposits, HMDA originations, or HMDA applications
  - Deposit shares yield appreciably higher levels of concentration
  - And mask a significant post-crisis increase in concentration



The Effect of Credit Concentration on Credit Supply

- Credit supply for different quartiles of credit concentration
  - Origination rates increase in response to monetary easing
  - <u>Increase is larger in less concentrated counties (bottom quartiles)</u>



Empirical Strategy: Credit Concentration

- Does local credit concentration affect the response of credit supply to monetary easing?
  - Dependent variable: Credit supply of lender *i* in county *j* in year *t*
  - Explanatory variable: Credit concentration in county *j* in year *t*

- Structural break approach with continuous treatment
  - Pre-treatment 2006-2007, post-treatment 2009-2010

$$S_{ijt} = \beta_0 \cdot Concentration_{jt-1} + \gamma_0 \cdot X_{ijt} + \delta_0 \cdot Y_{ijt} + \theta_0 \cdot Z_{jt} + L_i + C_j + T_t$$
$$+ \beta_1 \cdot Concentration_{jt-1} \cdot Post_t + \gamma_1 \cdot X_{ijt} \cdot Post_t + \delta_1 \cdot Y_{ijt} \cdot Post_t + \theta_1 \cdot Z_{jt} \cdot Post_t$$
$$+ L_i \cdot Post_t + \epsilon_{ijt}$$

  - Controls (X_{ijt}, Y_{ijt}, Z_{jt}), County FE (C_j), Lender FE (L_i), Year FE (T_t)
  - Controls x Post, Lender FE x Post
  - β_0 absorbs time-invariant cross-sectional biases
  - β_1 captures the post-treatment change in sensitivity, i.e. the causal effect

Empirical Strategy: Credit Market Power

- Concentration may correlate with unobserved market characteristics
- Solution: Examine within-county variation in credit market power
  - Explanatory variable: credit market power (square of credit market share) of lender *i* in county *j* in year *t*
  - County x Post FE to absorb time-varying county effects

$$
\begin{aligned}
S_{ijt} = {} & \beta_0 \cdot MarketPower_{ijt-1} + \gamma_0 \cdot X_{ijt} + \delta_0 \cdot Y_{ijt} + \theta_0 \cdot Z_{jt} + L_i + C_j + T_t \\
& + \beta_1 \cdot MarketPower_{ijt-1} \cdot Post_t + \gamma_1 \cdot X_{ijt} \cdot Post_t + \delta_1 \cdot Y_{ijt} \cdot Post_t + \theta_1 \cdot Z_{jt} \cdot Post_t \\
& + L_i \cdot Post_t + C_j \cdot Post_t + \epsilon_{ijt}
\end{aligned}
$$

Summary Statistics

Variable	2006-2007 mean	2006-2007 sd	2006-2007 25 pct	2006-2007 75 pct	2009-2010 mean	2009-2010 sd	2009-2010 25 pct	2009-2010 75 pct
Panel A: Dependent and explanatory variables								
Origination rate	0.60	0.24	0.46	0.78	0.73	0.18	0.67	0.85
Market concentration	0.02	0.01	0.02	0.03	0.06	0.02	0.05	0.07
Market power	0.001	0.002	0.000	0.001	0.008	0.014	0.000	0.010
Panel B: Lender-County controls (borrower characteristics)								
Average applicant income (000s)	108.994	50.469	78.450	128.125	120.971	51.448	93.315	137.478
Average loan amount (000s)	209.977	110.393	130.612	260.902	229.886	104.605	163.215	275.111
Average loan to income ratio	2.36	0.68	1.90	2.79	2.44	0.58	2.04	2.80
Share of jumbo loans	0.10	0.14	0.01	0.12	0.03	0.06	0.00	0.03
Share of male applicants	0.61	0.12	0.56	0.68	0.68	0.11	0.62	0.75
Share of female applicants	0.32	0.09	0.27	0.37	0.25	0.08	0.20	0.29
Share with applicant sex not reported	0.07	0.09	0.01	0.08	0.07	0.08	0.01	0.10
Share of minority applicants	0.26	0.21	0.08	0.39	0.08	0.11	0.02	0.10
Share of minority co-applicants	0.07	0.06	0.02	0.10	0.04	0.05	0.01	0.05
Share of owner-occupied units	0.89	0.11	0.86	0.96	0.91	0.10	0.89	0.97
Share of loans with first lien	0.82	0.16	0.73	0.94	0.97	0.07	0.98	1.00
Panel C: Lender-County controls (other)								
Log growth in number of applications	0.63	1.85	-0.18	0.42	0.85	1.83	-0.09	0.81
Share of originated loans sold	0.42	0.27	0.21	0.63	0.57	0.26	0.46	0.76
Share of bank portfolio in county	0.03	0.10	0.00	0.02	0.08	0.16	0.00	0.05
Median family income (000s)*	64.48	11.56	56.70	69.80	72.19	13.13	63.00	80.00
Ratio of county income to MSA mean*	1.09	0.16	0.99	1.17	1.24	0.20	1.11	1.36
Share of minority population*	0.32	0.21	0.16	0.46	0.22	0.15	0.10	0.30
Owner occupied units per capita*	0.26	0.04	0.23	0.29	0.28	0.03	0.26	0.30
Homes per capita*	0.33	0.05	0.30	0.36	0.34	0.05	0.32	0.37
Panel D: County controls								
Annual growth in HPI	0.05	0.07	0.01	0.09	-0.05	0.05	-0.08	-0.02
Unemployment rate	0.05	0.01	0.04	0.05	0.09	0.02	0.08	0.11

* Each lender-county-year observation is constructed by taking the mean over all applications' census-tract values

Main Results

1. Lenders' post-2008 increase in origination rates is lower in counties with higher credit concentration (column 1)
2. Within-county heterogeneity: increase in origination rates is lower for lenders with higher credit market power (column 2)

	Market Concentration (1)	Market Power (2)
Market Concentration	0.766***	
	(0.000)	
Market Concentration x Post	-0.575***	
	(0.000)	
Market Power		5.268***
		(0.000)
Market Power x Post		-4.324***
		(0.000)
Controls	YES	YES
LenderxPost, Year FE	YES	YES
County FE	YES	NO
CountyxPost FE	NO	YES
Number of observations	380,483	380,483
Adjusted R2	0.931	0.932

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

By Type of Loan Application

- The channel affects both loans for new purchases and for refinances

	Market Concentration		Market Power	
	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)
Market Concentration	0.861***	0.562***		
	(0.000)	(0.000)		
Market Concentration x Post	-0.653***	-0.483***		
	(0.000)	(0.000)		
Market Power			4.772***	5.081***
			(0.000)	(0.000)
Market Power x Post			-4.189***	-4.130***
			(0.000)	(0.000)
Controls	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES
County FE	YES	YES	NO	NO
CountyxPost FE	NO	NO	YES	YES
Number of observations	206,948	307,113	206,947	307,113
Adjusted R2	0.882	0.937	0.884	0.938

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Economic Significance Counterfactual

- Step 1
  - Reduce the post-crisis credit concentration down to some counterfactual threshold obtained from its empirical distribution across counties

	Empirical distribution for counterfactual	Threshold from empirical distribution
1	Post crisis (2009-2010)	lowest quartile
2	Post crisis (2009-2010)	minimum value
3	Pre crisis (2005)	lowest quartile
4	Pre crisis (2005)	minimum value

- Step 2
  - Use the empirical model to estimate the resulting aggregate increase in originations rates

DISTRIBUTION	Post crisis (2009-2010)		Pre crisis (2005)	
THRESHOLD	lowest quartile	minimum	lowest quartile	minimum
	(1)	(2)	(3)	(4)
Counterfactual increase (pps)	0.55	1.51	1.43	1.83
As ratio to actual increase (%)	4.25	11.56	10.98	14.05

The Deposits Channel (1)

- Unlike the deposit concentration channel, the credit concentration channel exerts significant local influence even after absorbing aggregate lender effects

	Market Concentration	
	Purchase	Refinance
	(1)	(2)
Market Concentration (Apps)	0.858***	0.557***
	(0.000)	(0.000)
Market Concentration (Apps) x Post	-0.661***	-0.482***
	(0.000)	(0.000)
Market Concentration (Deps)	0.030*	0.011
	(0.067)	(0.411)
Market Concentration (Deps) x Post	-0.004	-0.001
	(0.665)	(0.826)
Controls	YES	YES
LenderxPost, Year FE	YES	YES
County FE	YES	YES
CountyxPost FE	NO	NO
Number of observations	206,933	307,024
Adjusted R2	0.882	0.937

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

The Deposits Channel (2)

- The channel operates via both depository and non-depository institutions

	Market Concentration				Market Power			
	Depository		Non-Depository		Depository		Non-Depository	
	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Market Concentration	0.838***	0.582***	0.550**	0.572**				
	(0.000)	(0.000)	(0.042)	(0.033)				
Market Concentration x Post	-0.624***	-0.522***	-0.533**	-0.518**				
	(0.000)	(0.000)	(0.037)	(0.016)				
Market Power					4.898***	5.152***	5.954***	4.487***
					(0.000)	(0.000)	(0.000)	(0.001)
Market Power x Post					-4.238***	-4.135***	-6.272***	-3.165**
					(0.000)	(0.000)	(0.000)	(0.017)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	NO	NO	NO	NO
CountyxPost FE	NO	NO	NO	NO	YES	YES	YES	YES
Number of observations	174,774	253,611	32,134	53,483	174,773	253,611	32,017	53,418
Adjusted R2	0.878	0.931	0.915	0.971	0.880	0.933	0.915	0.971

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

More Cross-Section: Market Leaders vs Others

- The channel operates via both market leaders and laggards

	Market Concentration				Market Power			
	Top 5 Lenders		Other Lenders		Top 5 Lenders		Other Lenders	
	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Market Concentration	0.937***	0.679***	0.669***	0.238**				
	(0.001)	(0.002)	(0.000)	(0.025)				
Market Concentration x Post	-0.735***	-0.577***	-0.497***	-0.239**				
	(0.004)	(0.003)	(0.000)	(0.011)				
Market Power					3.038***	1.575***	20.393***	8.288***
					(0.000)	(0.000)	(0.000)	(0.000)
Market Power x Post					-2.377***	-0.993**	-17.114***	-5.162***
					(0.000)	(0.028)	(0.000)	(0.000)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	NO	NO	NO	NO
CountyxPost FE	NO	NO	NO	NO	YES	YES	YES	YES
Number of observations	15,882	20,333	132,696	207,582	15,882	20,333	132,696	207,582
Adjusted R2	0.864	0.945	0.898	0.941	0.866	0.947	0.898	0.942

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

More Cross-Section: Housing Crisis Counties vs Others

- The channel operates in counties regardless of their exposure to declines in housing prices

	Market Concentration				Market Power			
	LOW HPI DECLINE		HIGH HPI DECLINE		LOW HPI DECLINE		HIGH HPI DECLINE	
	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Market Concentration	0.770***	0.546***	0.849***	0.656***				
	(0.001)	(0.000)	(0.000)	(0.000)				
Market Concentration x Post	-0.556***	-0.247**	-0.664***	-0.633***				
	(0.001)	(0.016)	(0.000)	(0.000)				
Market Power					3.955***	3.650***	5.961***	5.507***
					(0.000)	(0.000)	(0.000)	(0.000)
Market Power x Post					-3.319***	-2.360***	-5.451***	-4.794***
					(0.000)	(0.000)	(0.000)	(0.000)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	NO	NO	NO	NO
CountyxPost FE	NO	NO	NO	NO	YES	YES	YES	YES
Number of observations	69,801	100,178	136,546	206,297	69,799	100,178	136,545	206,297
Adjusted R2	0.871	0.945	0.888	0.938	0.873	0.946	0.889	0.939

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Robustness: Using Pre-Crisis Concentration

- Results not driven by potentially endogenous within-county variation in concentration
  - Use pre-crisis concentration

	Market Concentration		Market Power	
	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)
Market Concentration	-	-		
Market Concentration x Post	-0.533***	-0.619***		
	(0.000)	(0.000)		
Market Power			4.485***	5.294***
			(0.000)	(0.000)
Market Power x Post			-1.817***	-3.262***
			(0.001)	(0.000)
Controls	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES
County FE	YES	YES	NO	NO
CountyxPost FE	NO	NO	YES	YES
Number of observations	206,948	307,113	206,947	307,113
Adjusted R2	0.882	0.937	0.884	0.938

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Robustness: The Aggregate Post-Crisis Shift in Concentration

- Results not driven by the post-crisis shift in credit concentration
  - Split the sample into low/high HHI growth counties

	Market Concentration				Market Power			
	LOW CHANGE IN HHI		HIGH CHANGE IN HHI		LOW CHANGE IN HHI		HIGH CHANGE IN HHI	
	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Market Concentration	0.650***	0.828***	0.669***	0.398**				
	(0.001)	(0.000)	(0.001)	(0.012)				
Market Concentration x Post	-0.528**	-0.452***	-0.527***	-0.424***				
	(0.011)	(0.009)	(0.001)	(0.001)				
Market Power					6.751***	5.091***	4.012***	4.544***
					(0.000)	(0.000)	(0.000)	(0.000)
Market Power x Post					-5.506***	-1.884***	-3.485***	-3.917***
					(0.000)	(0.001)	(0.000)	(0.000)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	NO	NO	NO	NO
CountyxPost FE	NO	NO	NO	NO	YES	YES	YES	YES
Number of observations	110,009	161,989	95,857	143,994	110,008	161,989	95,856	143,994
Adjusted R2	0.881	0.942	0.886	0.934	0.883	0.943	0.887	0.936

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Robustness: Excluding Entries + Balanced Sample

- Results not driven by endogenous entries/exits
  - Exclude entrants
  - Use balanced panel (small sample size)

	Market Concentration				Market Power			
	EXCLUDING ENTRANTS		BALANCED PANEL		EXCLUDING ENTRANTS		BALANCED PANEL	
	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Market Concentration	0.934***	0.615***	0.409*	0.561***				
	(0.000)	(0.000)	(0.097)	(0.000)				
Market Concentration x Post	-0.731***	-0.567***	-0.181	-0.494***				
	(0.000)	(0.000)	(0.443)	(0.000)				
Market Power					4.629***	4.998***	4.321***	4.293***
					(0.000)	(0.000)	(0.000)	(0.000)
Market Power x Post					-3.989***	-3.987***	-3.639***	-3.314***
					(0.000)	(0.000)	(0.000)	(0.000)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	NO	NO	NO	NO
CountyxPost FE	NO	NO	NO	NO	YES	YES	YES	YES
Number of observations	186,052	275,899	56,816	94,428	186,048	275,898	56,816	94,428
Adjusted R2	0.883	0.937	0.824	0.939	0.885	0.938	0.827	0.941

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Robustness: Alternative Measure of Concentration

- Results not driven by the choice of concentration index
  - Use the 5-firm concentration ratio

	Market Concentration		Market Power	
	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)
Market Concentration	0.112***	0.070***		
	(0.000)	(0.000)		
Market Concentration x Post	-0.078***	-0.049***		
	(0.000)	(0.001)		
Market Power			0.500***	0.479***
			(0.000)	(0.000)
Market Power x Post			-0.344***	-0.255***
			(0.000)	(0.000)
Controls	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES
County FE	YES	YES	NO	NO
CountyxPost FE	NO	NO	YES	YES
Number of observations	206,948	307,113	206,947	307,113
Adjusted R2	0.882	0.937	0.884	0.938

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Robustness: Alternative Measure of Credit Supply

- Results hold if supply is obtained from approval rates instead of origination rates

	Market Concentration		Market Power	
	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)
Market Concentration	0.868***	0.314***		
	(0.000)	(0.003)		
Market Concentration x Post	-0.759***	-0.475***		
	(0.000)	(0.000)		
Market Power			2.445***	3.644***
			(0.000)	(0.000)
Market Power x Post			-2.163***	-2.955***
			(0.000)	(0.000)
Controls	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES
County FE	YES	YES	NO	NO
CountyxPost FE	NO	NO	YES	YES
Number of observations	206,948	307,113	206,947	307,113
Adjusted R2	0.849	0.927	0.850	0.929

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Robustness: Alternative Definition of Local Market

- Results not driven by the unit of disaggregation
  - Define market at the census-tract level

	Market Concentration		Market Power	
	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)
Market Concentration	0.173***	0.388***		
	(0.000)	(0.000)		
Market Concentration x Post	-0.145***	-0.403***		
	(0.000)	(0.000)		
Market Power			0.770***	1.751***
			(0.000)	(0.000)
Market Power x Post			-0.617***	-1.249***
			(0.000)	(0.000)
Controls	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES
County FE	YES	YES	NO	NO
CountyxPost FE	NO	NO	YES	YES
Number of observations	308,033	966,831	307,982	966,823
Adjusted R2	0.738	0.845	0.738	0.847

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Robustness: Alternative Pre-Treatment Windows

- Results not driven by the choice of pre-treatment window
  - Make the pre-treatment window more narrow or more wide

	Market Concentration				Market Power			
	Excluding 2007		Including 2008		Excluding 2007		Including 2008	
	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance	Purchase	Refinance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Market Concentration	0.662***	0.735***	0.624***	0.466***				
	(0.001)	(0.000)	(0.000)	(0.000)				
Market Concentration x Post	-0.388**	-0.645***	-0.443***	-0.368***				
	(0.033)	(0.000)	(0.000)	(0.000)				
Market Power					3.322***	4.922***	3.630***	3.205***
					(0.000)	(0.000)	(0.000)	(0.000)
Market Power x Post					-2.739***	-3.970***	-3.047***	-2.254***
					(0.000)	(0.000)	(0.000)	(0.000)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
LenderxPost, Year FE	YES	YES	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	NO	NO	NO	NO
CountyxPost FE	NO	NO	NO	NO	YES	YES	YES	YES
Number of observations	141,005	218,327	247,737	367,053	140,997	218,325	247,736	367,053
Adjusted R2	0.897	0.941	0.868	0.935	0.899	0.942	0.869	0.936

p-values in parentheses, *** p<0.01, ** p<0.05, * p<0.10

Conclusion

- Credit concentration affects the transmission of monetary policy
- Channel is separate from the deposits channel
- Implications for distributive impact
- Thank you for your attention and comments!